Centralized Coal Logistics Management System

MINE → RAILWAY SIDING → POWER PLANT

CURRENT SCENARIO

Current Working Model

- Coal dispatch data maintained in **multiple Excel sheets**
- Separate files for:
  - ✓ Mine DPR
  - ✓ Siding loading
  - ✓ Rake-wise RR
  - ✓ Penalties & reconciliation

Key Issues

- No centralized visibility
- Manual reconciliation
- High dependency on individuals
- Delayed identification of penalties

CHALLENGES & RISKS

Operational Challenges

- Difficult to track coal movement end-to-end
- No real-time siding stock visibility
- Rake planning not linked with actual stock

Financial & Compliance Risks

- Demurrage & overloading penalties identified late
- Risk of data mismatch during audits
- Time-consuming management reporting

Excel-based control is no longer scalable or audit-safe

PROPOSED PROJECT

We are going to replace multiple Excel-based records with a single centralized web portal to manage the end-to-end coal movement from mines to railway sidings in Jharkhand and further to power plants in West Bengal.

1. Coal dispatch from mines by road (truck-wise DPR)
2. Receipt and stock management at railway sidings
3. Rake planning, wagon-wise loading, and weighment
4. Railway Receipt (RR), freight, GST, and other charges
5. Rail transit and power plant receipt
6. Automatic reconciliation, penalty calculation, and MIS reporting.

END-TO-END PROCESS FLOW

System Covers Entire Lifecycle

1. Mine dispatch by road (DPR)
2. Railway siding receipt, despatch and stock
3. Rake planning & demand
4. Wagon-wise loading & weighment
5. Railway Receipt (RR)
6. Rail transit
7. Power plant receipt
8. Reconciliation & penalties
9. MIS & dashboards

PENALTY CONTROL MECHANISM
(KEY BENEFIT)

Current Situation

- Penalties known **after RR received**
- No preventive control

Proposed System

- Live demurrage timer at siding
- Wagon overloading alerts before dispatch
- Auto-calculation based on railway rules

System helps in preventing penalties, not just recording them

RECONCILIATION & AUDIT READINESS

Automatic Reconciliation Between:

- Mine dispatch vs siding receipt
- Siding receipt vs rake loading
- Rake loading vs RR
- RR vs power plant receipt

Audit Advantages

- Single source of truth
- Locked data after RR
- Complete document trail
- User activity logs

MANAGEMENT DASHBOARD & MIS

Real-Time Visibility

- Daily & monthly dispatch
- Siding-wise performance
- Rake turnaround time
- Penalty trend analysis
- Plant-wise coal supply

Management Benefit

- Faster decisions
- Accurate reporting
- Zero manual compilation

BENEFITS

Tangible Benefits

- Reduction in demurrage & overloading penalties
- Faster reconciliation & billing
- Reduced manpower dependency

Intangible Benefits

- Better operational discipline
- Accountability at every level
- Scalable for future expansion